

CURRICULUM VITAE

PERSONAL INFORMATION



DUONG Ly Cao

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Gender Male | Date of birth 16/02/1995 | Nationality Vietnamese

I graduated from Ho Chi Minh City University of Technology and subsequently obtained a Master's degree from Pukyong National University in Busan, South Korea, specializing in Electrical and Biomedical Engineering. My academic background is strongly grounded in signal and image processing, with a particular focus on automation systems and medical devices. I am currently a PhD researcher in the Medical Physics department at Carl von Ossietzky Universität Oldenburg, where my research focuses on AI-based solutions for hearing acoustics.

EDUCATION AND RESEARCHING

From November 2025
to now

Carl von Ossietzky Universität Oldenburg, Germany

PhD Researcher in Medical Physics, AI-based solutions in Acoustics

From September 2021
to November 2025

Ohlabs - Pukyong National University, Busan, Republic of Korea

Cooperation Researcher

Publications:

- Fcg-Former: Identification of functional groups in FTIR spectra using enhanced transformer-based model. Analytical Chemistry (2024) – Under review
- Smart Low Level Laser Therapy System for Automatic Facial Dermatological Disorder Diagnosis. IEEE Journal of Biomedical and Health Informatics (2023)

From September 2018
to September 2021

Pukyong National University, Busan, Republic of Korea

Master's degree in Electrical and Biomedical Engineering

Graduated date: 08/2020

GPA: 4.25 / 4.5

Publications:

- Full-view in vivo Skin and Blood Vessels Profile Segmentation in Photoacoustic Imaging based on DeepLearning. Photoacoustic (2022)

- Development of fast photoacoustic and ultrasound imaging system based on slider-crank scanner for small animals and humans study. Expert Systems with Applications (2022)

From August 2013 to
April 2018

Ho Chi Minh City University of Technology

Bachelor's degree in Biomedical Engineering
GPA: 8.0 / 10.0

Publications:

- Model of colposcopy using polarized light and effective early, *ISEPM*, Ho Chi Minh, 2017.

WORK EXPERIENCE

From April 2025 to
November 2025

Researcher at Inha University Industry-Academic Cooperation Foundation.

Responsibility:

AI-driven solutions for MRI-based medical imaging analysis. Designed and implemented AI algorithms for automated MRI interpretation, with emphasis on 3D segmentation of lumbar and ankle structures.

From June 2022 to
April 2025

Senior AI Researcher at Vision-in Inc.

Responsibility:

My professional responsibilities encompass research, development, and deployment of deep learning models in the field of Surveillance. Additionally, I provide support to the platform team in optimizing model inference, improving processing time, and increasing the number of channels on multi-platform environments.

Projects:

- Fire and smoke synthetic images generation based on Diffusion
- Human searching based on image description model
- Fire, smoke, and human actions detection with limited datasets enhanced with contrastive learning pretrained model
- Developing cross-platform plugins for NX Witness
- Optimizing resources by deploying models and platform on Edge devices (NVIDIA Jetson board)

From October 2021 to
June 2022

AI Research Engineer at AIDOT Inc.

Responsibility:

Developing deep learning models (detection, classification, segmentation) for ultrasound Carotid Artery diseases. Optimizing and Quantization models on tablet devices.

TECHNICAL SKILL

Programming Languages

Python

C++

JavaScript

9/10

8/10

6/10

AI Tools/ Platforms

Pytorch

ONNX Runtimes

Tensorflow

8/10

8/10

7/10

Others

VTK Toolkit

NX Witness Plugin

Docker

	7/10	8/10	7/10
PERSONAL QUALITIES			
Languages	Vietnamese	English	Korean
	Native	Advanced	Basic
	• High motivated and eager to learn new things • Good at interpersonal skills with an ability to collaborate well and work across team • Fast self-learner and team player		
ADDITIONAL INFORMATION			
Project	Facial Recognition Responsibility: Building a facial recognition API server for a check-in/check-out project in Vietnam, where we control user IDs, enroll users' faces, and verify them using a tablet camera device.		
	3D Photoacoustic Visualize Software Responsibility: Developing a 3D viewer software for photoacoustic imaging with VTK (Visualization Toolkit) engine can provide a versatile and powerful visualization tool for analyzing photoacoustic imaging data. Creating software that supports various medical data formats including DICOM, NRRD, and TDMS.		
ACTIVITIES			
	Volunteer of the Green Summer Volunteer Campaign 2014 in Tra Vinh province, help the people build 1800m concrete roads through their village Volunteer of the National University Entrance Exam Support Campaign 2016, disseminating information, providing information about the examination to examinees, introducing free and cheap motels, restaurant and more		
HONORS AND AWARDS			
	Study encouragement scholarship of 131, 141, 151, 172 semester, Bach Khoa University Incentive award The 2017 Eureka Student Research Competition with project “Model of colposcopy using polarized light and effective early”		